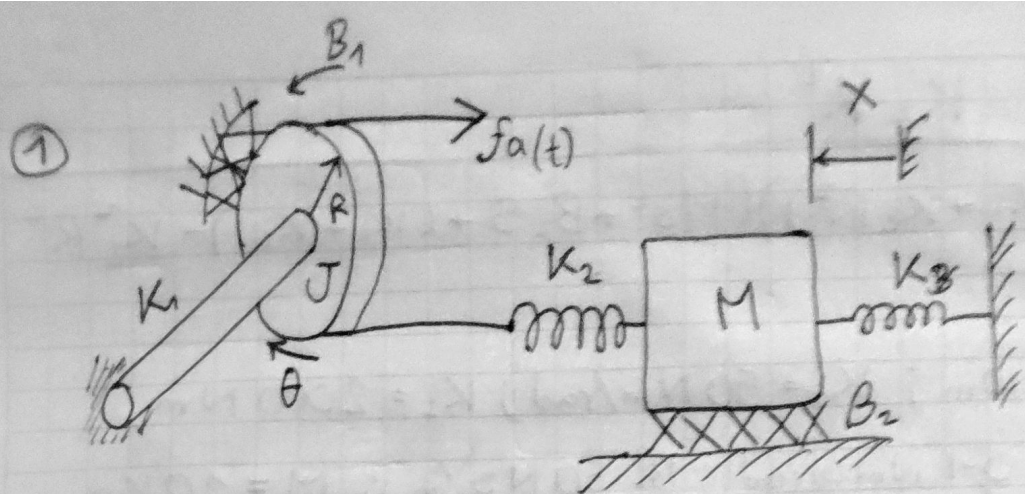




$$\theta(t) = q_1$$

$$x(t) = q_2$$

$$\dot{\theta}(t) = \dot{q}_1$$

$$\dot{x}(t) = \dot{q}_2$$

$$T = \frac{1}{2} J \dot{\theta}^2 + \frac{1}{2} M \dot{x}^2$$

$$V = \frac{1}{2} K_1 \theta^2 + \frac{1}{2} K_2 (R\theta - x)^2 + \frac{1}{2} K_3 x^2$$

$$D = \frac{1}{2} B_1 \dot{\theta}^2 + \frac{1}{2} B_2 \dot{x}^2$$

Euler-Lagrange con Disipación

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} + \frac{\partial D}{\partial \dot{q}_i} = Q_i$$

$$\theta: J \ddot{\theta} + B_1 \dot{\theta} + (K_1 + K_2 R^2) \theta - K_2 R x = R f_a(t)$$

$$x: M \ddot{x} + B_2 \dot{x} - K_2 R \theta + (K_2 + K_3) x = 0$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_1 + K_2 R^2}{J} & -\frac{B_1}{J} & \frac{K_2 R}{J} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{K_2 R}{M} & 0 & -\frac{K_2 + K_3}{M} & -\frac{B_2}{M} \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ \frac{R}{J} \\ 0 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad D = 0$$

$$\frac{X(s)}{F_a(s)} = \frac{K_2 R^2}{(J s^2 + B_1 s + (K_1 + K_2 R^2))(M s^2 + B_2 s + (K_2 + K_3)) - K_2^2 R^2}$$

$$J = 0,05 \text{ Kg m}^2; R = 0,10 \text{ m}; K_1 = 50 \text{ Nm/rad}; K_2 = 2000 \text{ Nm};$$

$$K_3 = 1000 \text{ N/m}; B_1 = 0,1 \text{ Nm s/rad}; B_2 = 50 \text{ N s/m}; M = 10 \text{ Kg}$$

$$\frac{X(s)}{F_a(s)} = \frac{40}{s^4 + 7s^3 + 1710s^2 + 7600s + 340000}$$

$$\triangle \quad \text{Tipo } 0 \rightarrow 1$$

$$s^4 \rightarrow s^5 = V$$

DDDDPI

$$ess = 0, \quad \square$$

$$t_{sla} = 5 \text{ s} \quad 3d = 1$$

$$t_{sd} = 4,5 \text{ s}$$

$$\omega_n = \frac{4}{4,5(1 - \sqrt{1 - 0,88^2})} = 0,88$$

$$PIDDD^3 = K_p + \frac{K_i}{s} + K_{d1}s + K_{d2}s^2 + K_{d3}s^3$$

$$= \frac{K_{d3}s^4 + K_{d2}s^3 + K_{d1}s^2 + K_p s + K_i}{s}$$

$$\overset{1}{s^5} + \overset{\checkmark}{7s^4} + \overset{\Delta}{1710s^3} + \overset{\square}{7600s^2} + \overset{1}{340000s} + \overset{\checkmark}{40K_{d3}s^4} + \overset{\checkmark}{40K_{d2}s^3} + \overset{\nabla}{40K_{d1}s^2}$$

$$+ \overset{\square}{40K_p s} + 40K_i = P.D.$$

$$s^5 + s^4(7 + 40K_{d3}) + s^3(1710 + 40K_{d2}) + s^2(7600 + 40K_{d1}) + s(340000 + 40K_p) + 40K_i = P.D.$$

$$P.D. = (s + 2(1)(0,88)s + 0,88^2)(s + 10(1)(0,88))^3$$

$$= s^5 + 28,16s^4 + 279,6s^3 + 1111s^2 + 137900s + 527,7$$

$$28,16 = 7 + 40K_{d3}; \quad 1379 = 340000 + 40K_p$$

$$279,6 = 1710 + 40K_{d2}; \quad 527,7 = 40K_i$$

$$1111 = 7600 + 40K_{d1};$$

$$K_{d1} = -162,225$$

$$K_{d2} = -35,76$$

$$K_{d3} = 0,529$$

$$K_p = -8465,52$$

$$K_i = 13,19$$

$$F(s) = \frac{40}{s^4 + 7s^3 + 1710s^2 + 7600s + 340000}$$

$$s_1 = \overbrace{-1,08}^{\alpha} \pm \overbrace{38,28}^{\omega}$$

$$s_2 = -2,41 \pm 15,025$$

$$\frac{I_{nd}/I_{nd} (\alpha^2 + \omega^2)}{s^2 + (-2\alpha)s + (\alpha^2 + \omega^2)}$$

$$F(s) = \frac{40/340000 (-1,08^2 + 38,28^2)}{s^2 + (-2(-1,08))s + (-1,08^2 + 38,28^2)}$$

$$F(s) = \frac{0,1727}{s^2 + 2,164s + 1467,91}$$

$$e_{ss} = 0; \quad \checkmark$$

$$\zeta_d = \cancel{1} 1$$

$$t_{sd} = 4,5$$

$$\text{Tipo } 0 \rightarrow 1 = 1I$$

$$\omega_n = \cancel{0,88} 0,88$$

$$s^2 \rightarrow s^3 = PID = K_p + \frac{K_i}{s} + K_d s$$

$$= \frac{K_d s^2 + K_p s + K_i}{s}$$

$$s^3 + 2,164 s^2 + 1467,91 s + 0,1727 K_d s^2 + 0,1727 K_p s + 0,1727 K_i = P.D.$$

$$s^3 + s^2(2,164 + 0,1727 K_d) + s(1467,91 + 0,1727 K_p) + 0,1727 K_i = P.D.$$

$$(s + 2(-1)(0,88)s + 0,88^2)(s + 10(-1)(0,88))$$

$$s^3 + 10,56 s^2 + 16,26 s + 6,815 = P.D.$$

$$10,56 = 2,164 + 0,1727 K_d \quad ; \quad K_d = 48,616$$

$$16,26 = 1467,91 + 0,1727 K_p \quad ; \quad K_p = -8405,64$$

$$6,815 = 0,1727 K_i \quad ; \quad K_i = 39,46$$

$$\zeta = 1; \quad 0,4 \leq \omega_n \cdot T \leq 1,75 \quad ; \quad \omega_n = 0,88$$

$$0,455 \leq T \leq 1,988$$

$$\underline{T = 0,5}$$

$$S = \frac{1 - z^{-1}}{T}$$

$$\frac{K_d S^2 + K_p S + K_i}{S} = \frac{\left(K_d \left(\frac{1-\bar{z}^{-1}}{0,5} \right)^2 + K_p \left(\frac{1-\bar{z}^{-1}}{0,5} \right) + K_i \right) \cdot 0,5}{\left(\frac{1-\bar{z}^{-1}}{0,5} \right) \cdot 0,5}$$

$$\frac{(K_d/0,5)(1-\bar{z}^{-1})^2 + K_p(1-\bar{z}^{-1}) + K_i(0,5)}{1-\bar{z}^{-1}} = \frac{K_d/0,5(1-2\bar{z}^{-1}+\bar{z}^{-2}) + K_p(1-\bar{z}^{-1}) + K_i \cdot 0,5}{1-\bar{z}^{-1}}$$

$$\frac{K_d/0,5 - 2K_d \bar{z}^{-1} + K_d \bar{z}^{-2} + K_p - K_p \bar{z}^{-1} + K_i \cdot 0,5}{1-\bar{z}^{-1}}$$

$$\frac{\left(\frac{K_d}{0,5} + K_p + K_i \cdot 0,5 \right) + \left(-K_p - \frac{2K_d}{0,5} \right) \bar{z}^{-1} + \left(\frac{K_d}{0,5} \right) \bar{z}^{-2}}{1-\bar{z}^{-1}}$$

$$\left. \begin{array}{l} q_2 = 97,232 \\ q_1 = 8211,146 \\ q_0 = -8288,648 \end{array} \right\} 0,5 = T \quad \left. \begin{array}{l} q_2 = 106,84 \\ q_1 = 8191,9 \\ q_0 = -8280,8 \end{array} \right\} 0,455 = T$$

$$\frac{0,1727}{S^2 + 2,164S + 1467,91} \rightarrow \frac{0,1727}{S(S^2 + 2,164S + 1467,91)}$$

$$S = -1,082 \pm 38,29j$$

C.C.

$$t_s = 4,5s$$

$$\frac{A}{S} + \frac{B(S+1,082)}{(S+1,082)^2 + 38,29^2} + \frac{C(38,29)}{(S+1,082)^2 + 38,29^2}$$

$$T = \frac{45}{50} = 0,9$$

$$3 < 0,7$$

$$\frac{0,00022 \bar{z} + 0,000197 \bar{z}^{-1}}{\bar{z}^2 + 1,731 \bar{z} + 0,823 \bar{z}^{-1}} \rightarrow \frac{0,00022 + 0,000197 \bar{z}^{-1}}{1 - 1,731 \bar{z}^{-1} + 0,823 \bar{z}^{-2}}$$

$$ess = 0$$

$$C_{over} = -1,116$$

$$\frac{0,00022 + 0,000197 \bar{z}^{-1}}{1 - 1,731 \bar{z}^{-1} + 0,823 \bar{z}^{-2}} \left(\frac{r_0 + r_1 \bar{z}^{-1} + r_2 \bar{z}^{-2}}{(1-\bar{z}^{-1})} \right)$$

$$P_{oln} = -0,865 \pm 0,272j$$

$$T_{yer} 0 \rightarrow 1 ; 1I$$

$$\begin{aligned}
 & (1 - 1,731 \bar{z}^1 + 0,823 \bar{z}^2)(1 - \bar{z}^1) + (0,00022 + 0,000197 \bar{z}^1)(r_0 + r_1 \bar{z}^1 + r_2 \bar{z}^2) \\
 & 1 - 1,731 \bar{z}^1 + 0,823 \bar{z}^2 - \bar{z}^1 + 1,731 \bar{z}^2 - 0,823 \bar{z}^3 + 0,00022 r_0 + 0,00022 r_1 \bar{z}^1 \\
 & + 0,00022 r_2 \bar{z}^2 + 0,000197 r_0 \bar{z}^1 + 0,000197 r_1 \bar{z}^2 + 0,00022 r_2 \bar{z}^3 \\
 & 1,00022 \bar{z}^1 (-1,731 - 1 + 0,00022 r_1 + 0,000197 r_0) + \bar{z}^2 (0,823 + 1,731 + 0,00022 r_2) \\
 & + \bar{z}^3 (-0,823 + 0,00022 r_2)
 \end{aligned}$$

$$\zeta < 0,7$$

Sub. Am

$$T = 0,09$$

$$PD = (1 + p_1 \bar{z}^1 + p_2 \bar{z}^2)(1 - 0,1 \bar{z}^1)$$

$$p_1 = -2 e^{-\zeta \omega_n T} \cos(\omega_n T \sqrt{1 - \zeta^2})$$

$$p_2 = e^{-2\zeta \omega_n T}$$

$$p_1 = -1,666$$

$$p_2 = 0,748$$

$$z = -0,865 \pm 0,272j$$

$$|z| = 0,865 = e^{-T \omega_n \zeta}$$

$$\angle = 0,272 = T \omega_n \sqrt{1 - \zeta^2}$$

$$\omega_n = \frac{0,272}{T \sqrt{1 - \zeta^2}}$$

$$(1 - 1,666 \bar{z}^1 + 0,748 \bar{z}^2)(1 - 0,1 \bar{z}^1)$$

$$1 - 1,666 \bar{z}^1 + 0,748 \bar{z}^2 - 0,1 \bar{z}^1 + 0,166 \bar{z}^2$$

$$-0,0748 \bar{z}^3 = P.D. \quad 0,00748 \bar{z}^3 = P.D.$$

$$\zeta = 0,470$$

$$\omega_n = 3,424$$

$$1 - \bar{z}^1(1,766) + \bar{z}^2(0,914) - 0,0748 \bar{z}^3 = P.D.$$

$$1,00022 r_0 = 1$$

$$r_0 = 0,999$$

$$-2,731 + 0,00022 r_1 + 0,000197 r_0 = 1,766$$

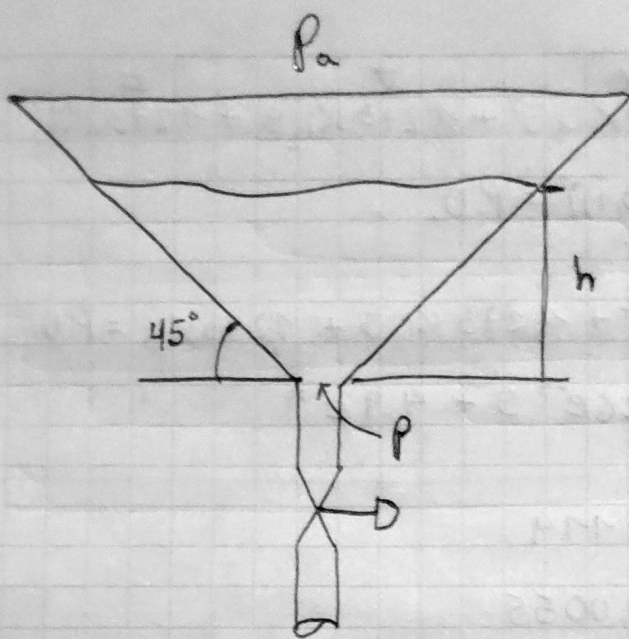
$$r_1 = 20440,01$$

$$2,554 + 0,00022 r_2 = 0,914$$

$$r_2 = -7454,54$$

$$-0,823 + 0,00022 r_2 = -0,0748$$

$$r_2 = 3400,90$$



$$r(h) = h \tan(45^\circ) = h$$

$$A(h) = \pi r(h)^2 = \pi h^2$$

$$V(h) = \frac{1}{3} \pi r(h)^2 h = \frac{1}{3} \pi h^3$$

$$\frac{dV}{dh} = \pi h^2 = A(h)$$

$$\frac{dV}{dt} = -Q(t)$$

$$P = P_a + \rho g h$$

$$\pi h^2 \frac{dh}{dt} = -Q(t) \rightarrow \frac{dh}{dt} = -\frac{Q(t)}{\pi h^2} \rightarrow -\frac{C_v}{\pi h^2} \sqrt{\rho g h}$$

$$Q = C_v \sqrt{\Delta P}$$

$$\Delta P = P - P_a = \rho g h$$

$$Q = C_v \sqrt{\rho g h}$$

$$\frac{dh}{dt} = -\frac{C_v \sqrt{\rho g}}{\pi h^{3/2}}$$

$$F(s) = \frac{h_0^{-2}/\pi}{s + \frac{K}{2\pi} h_0^{-5/2}} =$$

$$\frac{1,273}{s + 0,0089}$$

$$t_{sla} = 600 \text{ s}$$

$$t_{sd} = 550 \text{ s}$$

$$\zeta_d = 0,7$$

$$\omega_n = 0,010$$

$$\rho = 1000 \text{ Kg/m}^3 \quad C_v = 1 \times 10^{-4}$$

$$g = 9,806 \text{ m/s}^2$$

$$h_0 = 0,50 \text{ m}$$

$$K = C_v \sqrt{\rho g} = 0,0099$$

$$e_{ss} = 0 ; \quad \curvearrowright$$

$$T_{po} 0 \rightarrow 3 = 3I$$

$$s \rightarrow s^4$$

$$P I I^3 I^3$$

$$K_p + \frac{K_{i1}}{s} + \frac{K_{i2}}{s^2} + \frac{K_{i3}}{s^3}$$

$$= \frac{K_p s^3 + K_{i1} s^2 + K_{i2} s + K_{i3}}{s^3}$$

$$S^4 + 0,0089 S^3 + 1,273 K_p S^3 + 1,273 K_{i1} S^2 + 1,273 K_{i2} S + 1,273 K_{i3} = P.D.$$

$$(S + 2(0,7)(0,01)S + 0,01^2)(S + 10(0,7)(0,1))^2 = P.D.$$

$$S^4 + S^3(0,0089 + 1,273 K_p) + 1,273 K_{i1} S^2 + 1,273 K_{i2} S + 1,273 K_{i3} = P.D.$$

$$S^4 + 0,154 S^3 + 0,00696 S^2 + 8,26 e^{-5} S + 4,9 e^{-7} = P.D.$$

$$0,154 = 0,0089 + 1,273 K_p; \quad K_p = 0,114$$

$$0,00696 = 1,273 K_{i1}; \quad K_{i1} = 0,0055$$

$$8,26 e^{-5} = 1,273 K_{i2}; \quad K_{i2} = 0,000065$$

$$4,9 e^{-7} = 1,273 K_{i3}; \quad K_{i3} = 0,00000385$$

$$\zeta = 0,7; \quad 0,25 \leq \omega_n \cdot T \leq 1; \quad \omega_n = 0,010$$

$$25 \leq T \leq 100; \quad T = 30$$

$$\frac{K_p S^3 + K_{i1} S^2 + K_{i2} S + K_{i3}}{S^3} = S = \frac{1 - \bar{z}^{-1}}{T}$$

$$\left(K_p \left(\frac{(1 - \bar{z}^{-1})^3}{T^3} \right) + K_{i1} \left(\frac{(1 - \bar{z}^{-1})^2}{T^2} \right) + K_{i2} \left(\frac{1 - \bar{z}^{-1}}{T} \right) + K_{i3} \right) * T^3$$

$$= \frac{(1 - \bar{z}^{-1})^3}{T^3} * T^3$$

$$K_p(1 - 3\bar{z}^{-1} + 3\bar{z}^{-2} - \bar{z}^{-3}) + K_{i1}T(1 - 2\bar{z}^{-1} + \bar{z}^{-2}) + K_{i2}T^2(1 - \bar{z}^{-1}) + K_{i3}T^3$$

$$1 - 3\bar{z}^{-1} + 3\bar{z}^{-2} - \bar{z}^{-3}$$

$$\frac{\overset{q_0}{(K_p + K_{i1}T + K_{i2}T^2 + K_{i3}T^3)} + \overset{q_1}{\bar{z}^{-1}(-3K_p - 2K_{i1}T - K_{i2}T^2)} + \overset{q_2}{\bar{z}^{-2}(3K_p + K_{i1}T)} + \overset{q_3}{\bar{z}^{-3}K_p}}{1 - 3\bar{z}^{-1} + 3\bar{z}^{-2} - \bar{z}^{-3}}$$

$$q_0 = 0,348$$

$$q_1 = -0,7305$$

$$q_2 = 0,5070$$

$$q_3 = 0,114$$

$$30 = T$$

$$\frac{1,273}{s + 0,0089}$$

$$s = -0,0089$$

$$0,7 \leq z \leq 1$$

$$t_{sd} = 550$$

$$T = \frac{550}{50} = 11,5$$

$$\frac{1,273}{s(s + 0,0089)} = \frac{A(s + 0,0089)}{s} + \frac{B(s)}{s + 0,0089}$$

$$A + B = 0 \quad ; \quad A = 143,03$$

$$A(0,0089) = 1,273 \quad ; \quad B = -143,03$$

$$\frac{143}{s} + \frac{(-143)}{s + 0,0089} \xrightarrow{z} \frac{143 \cdot z}{z - 1} + \frac{(-143) \cdot z}{z - 0,906}$$

$$\frac{143z^2 - 129,55 - 143z^2 + 143}{(z - 1)(z - 0,906)} = \frac{13,442}{(z - 1)(z - 0,906)}$$

ade

$$\frac{143}{1 - z^{-1}} + \frac{(-143)}{1 - 0,906z^{-1}} = \frac{143 - 129,55z^{-1} - 143 + 143z^{-1}}{(1 - z^{-1})(1 - 0,906z^{-1})}$$

$$\frac{13,442z^{-1}}{(1 - z^{-1})(1 - 0,906z^{-1})} (1 - z^{-1}) = \frac{13,442z^{-1}}{1 - 0,906z^{-1}}$$

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